Machine Learning

AIML CZG565

M8 : Bayesian Learning

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Contact Session 10,11

Bayesian Learning
Parameter Estimation

- Where does the **cost** come from? - Logistic regression
- Why least-squares **cost** function, be a reasonable choice? – Linear regression

Distribution

Mileage (in kmpl)	*Car Price (in cr)*
9.8	10.48
9.12	1.75
9.5	6.95
10	2.51

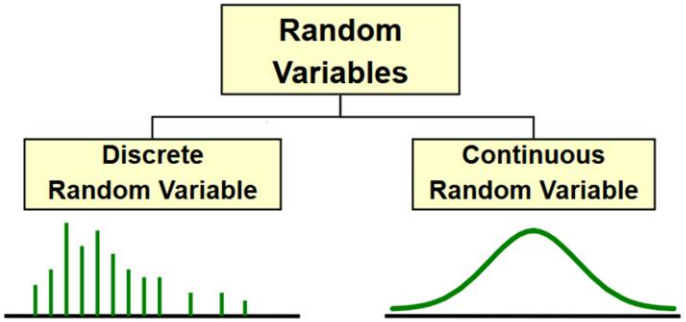
Mileage (in kmpl)	*Car Price (in cr)*
Neutral	High
Less	Low
Neutral	Medium
More	Low

Mileage (in kmpl)	*Car Price (in cr)*
9.8	High
9.12	Low
9.5	High
10	Low

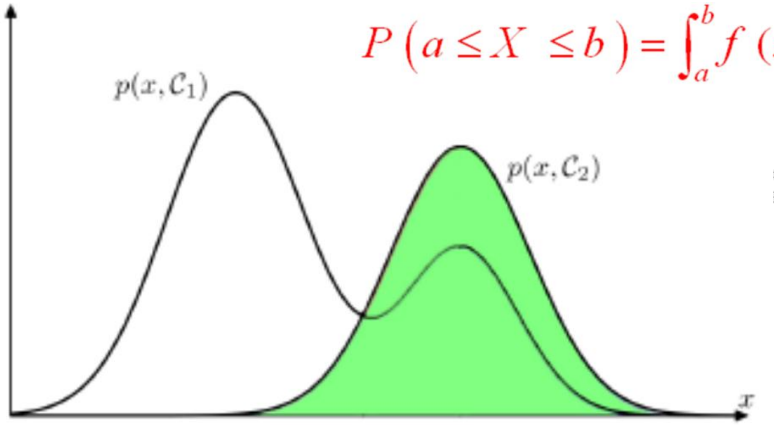
$$p(X = x_i, Y = y_j) = \frac{n_{ij}}{N}$$

$$p(Y = y_j | X = x_i) = \frac{n_{ij}}{c_i}$$

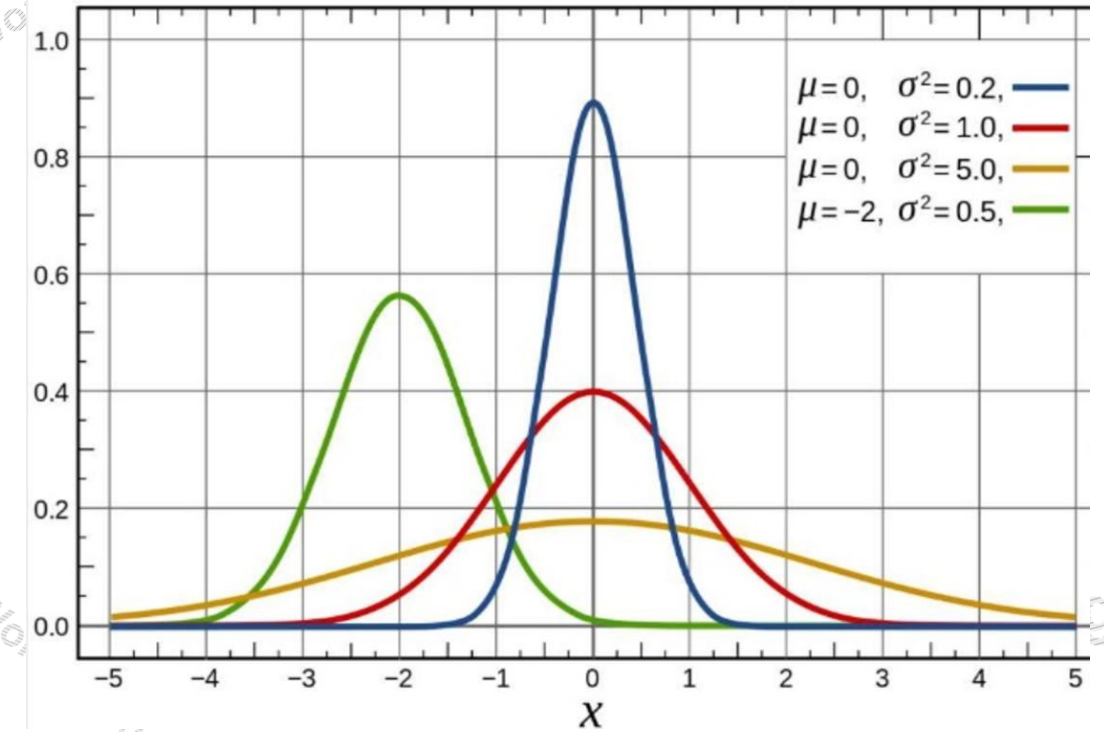
Represents a possible numerical value from a random event



$$P(a \le X \le b) = \int_a^b f(x)dx$$



Parameter Estimation



Distribution	Parameters
Bernoulli(p)	$\theta = p$
Poisson(λ)	$\theta = \lambda$
Uniform(a,b)	$\theta = (a,b)$
Normal(μ,σ^2)	$\theta = (\mu,\sigma^2)$
Y = mX + b	$\theta = (m,b)$

θ is the parameter of a distribution.

θ can be a vector of parameters

Distribution = model + parameter θ

$$N(x \mid \mu,\sigma^2) = \frac{1}{(2\pi\sigma^2)^{1/2}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

Find $\boldsymbol{\theta} = (\mathbf{Mean}, \mathbf{SD})$ from the data $\mathbf{X_i}$

ie., (x_i, $P(x_i)$)

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Parameter in ML

Mileage (in kmpl)	*Car Price (in cr)*
9.8	10.48
9.12	1.75
9.5	6.95
10	2.51

$$\text{CarPrice} = 8.5 + 0.5\ \text{Mileage} - 1.5\ \text{Mileage}^2$$

$$\text{Parameters}: (\theta_0, \theta_1, \theta_2)$$

Mileage (in kmpl)	*Car Price (in cr)*
9.8	High
9.12	Low
9.5	High
10	Low

$$\text{CarPrice} = \frac{1}{1+e^{-8.5 + 0.5\ \text{Mileage} - 1.5\ \text{Mileage}^2}}$$

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Parameter Estimation in ML

Mileage (in kmpl)	*Car Price (in cr)*
9.8	10.48
9.12	1.75
9.5	6.95
10	2.51

CarPrice = 8.5 + 0.5 Mileage – 1.5 Mileage^2

$y^{(i)} = \theta^T x^{(i)} + \varepsilon^{(i)}$

$\varepsilon^{(i)} \sim N(0, \sigma^2)$

Parameters : $(\theta_0, \theta_1, \theta_2)$

Find $\boldsymbol{\theta} = (\boldsymbol{\theta_0}, \boldsymbol{\theta_1}, \boldsymbol{\theta_2})$ from the data $\mathbf{X_i}$

ie., (Mileage$_i$, CarPrice$_i$)

Assumption: Data are__

- IID samples: $X_1 \ldots X_n$ where all X_i are independent and have the same distribution.
- Either same PMF (discrete) or same PDF (continuous)
- f (X | θ)
  Likelihood of different values of X depends on the values of our parameters θ
  f( ) is either PDF or PMF

Maximum Likelihood Estimation (MLE)

select that parameters θ that make the observed data the most likely

$f(X_1, X_2 \ldots X_n \mid \boldsymbol{\theta})$

Maximum A Posteriori (MAP)

choose the parameters θ that is the most likely, given the data

$f(\boldsymbol{\theta} \mid X_1, X_2 \ldots X_n)$

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Intuition of Bayes Theorem

MAP : f (θ | X_1 , X_2 .....X_n)

MLE: f (X_1 , X_2 .....X_n | θ)

$$P(h|D) = \frac{P(D|h)P(h)}{P(D)}$$

P(h) = prior probability of hypothesis h

P(D) = prior probability of training data D

P(h|D) = probability of h given D

P(D|h) = probability of D given h

$L(\theta)$, probability of data given parameter θ

After seeing data, posterior belief of θ

posterior likelihood prior

$$P(\theta|\text{data}) = \frac{P(\text{data}|\theta)P(\theta)}{P(\text{data})}$$

Before seeing data, prior belief of θ e.g. what is distribution over parameters θ

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•Maximum Likelihood Estimation (MLE)

1. Determine formula for $LL(\theta)$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ

3. Solve

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MLE – Linear Regression Model

Given the noise $\varepsilon^{(i)}$ obeys a Normal distribution each $y^{(i)}$ must also obey a Normal distribution around the true target value

Mileage (in kmpl)	*Car Price (in cr)*
9.8	10.48
9.12	1.75
9.5	6.95
10	2.51

CarPrice = 8.5 + 0.5 Mileage – 1.5 Mileage^2

$y^{(i)} = \theta^T x^{(i)} + \varepsilon^{(i)}$

$\varepsilon^{(i)} \sim N(0, \sigma^2)$

$$p(\epsilon^{(i)}) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(\epsilon^{(i)})^2}{2\sigma^2}\right).$$

Parameters : $(\theta_0, \theta_1, \theta_2)$

Find $\boldsymbol{\theta} = (\boldsymbol{\theta_0}, \boldsymbol{\theta_1}, \boldsymbol{\theta_2})$ from the data $\mathbf{X_i}$

ie., (Mileage_i, CarPrice_i)

$y^{(i)} \mid x^{(i)}; \theta \sim N(\theta^T x^{(i)}, \sigma^2)$

Maximum Likelihood Estimation (MLE)

select that parameters θ that make the observed data the most likely

$f(X_1, X_2 \ldots\ldots X_n \mid \boldsymbol{\theta})$

$$p(y^{(i)}|x^{(i)};\theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2}\right)$$

•MLE – Linear Regression Model

$$p(y^{(i)}|x^{(i)};\theta) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{(y^{(i)}-\theta^T x^{(i)})^2}{2\sigma^2}\right)$$

1. Determine formula for $LL(\theta)$

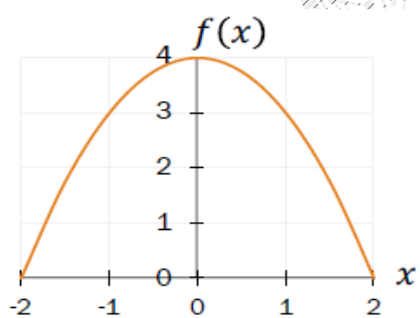
$$L(\theta) = \prod_{i=1}^{n} p(y^{(i)} \mid x^{(i)};\theta)$$

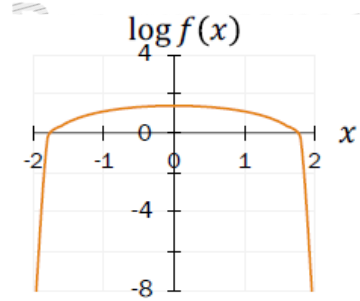
$$= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{(y^{(i)}-\theta^T x^{(i)})^2}{2\sigma^2}\right)$$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ

$$\hat{\theta} = \underset{\theta}{\operatorname{argmax}}\, L(\theta) \qquad LL(\theta) = \log L(\theta)$$

3. Solve





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MLE answers the question: For which parameter value does the observed data have the largest probability?

•MLE – Linear Regression Model

$$p(y^{(i)}|x^{(i)};\theta) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{(y^{(i)}-\theta^T x^{(i)})^2}{2\sigma^2}\right)$$

1. Determine formula for $LL(\theta)$

$$L(\theta) = \prod_{i=1}^{n} p(y^{(i)} \mid x^{(i)};\theta)$$

$$= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{(y^{(i)}-\theta^T x^{(i)})^2}{2\sigma^2}\right)$$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ

$$\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}}\, L(\boldsymbol{\theta}) \qquad LL(\theta) = \log L(\theta)$$

3. Solve

$$= \log L(\theta)$$

$$= \log \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{(y^{(i)}-\theta^T x^{(i)})^2}{2\sigma^2}\right)$$

$$= \sum_{i=1}^{n} \log \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{(y^{(i)}-\theta^T x^{(i)})^2}{2\sigma^2}\right)$$

$$= n\log\frac{1}{\sqrt{2\pi}\sigma} - \frac{1}{\sigma^2}\cdot\frac{1}{2}\sum_{i=1}^{n}(y^{(i)}-\theta^T x^{(i)})^2$$

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•MLE – Linear Regression Model

$$p(y^{(i)}|x^{(i)};\theta) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{(y^{(i)}-\theta^T x^{(i)})^2}{2\sigma^2}\right)$$

1. Determine formula for $LL(\theta)$

$$\log L(\theta) = n\log\frac{1}{\sqrt{2\pi}\sigma} - \frac{1}{\sigma^2}\cdot\frac{1}{2}\sum_{i=1}^{n}(y^{(i)}-\theta^T x^{(i)})^2$$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ

$$= -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y^{(i)}-\theta^T x^{(i)})^2$$

$$= -\frac{1}{2}\sum_{i=1}^{n}(y^{(i)}-\theta^T x^{(i)})^2$$

3. Solve

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•MLE – Linear Regression Model

$$p(y^{(i)}|x^{(i)};\theta) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{(y^{(i)}-\theta^T x^{(i)})^2}{2\sigma^2}\right)$$

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$$\log L(\theta) = n\log\frac{1}{\sqrt{2\pi}\sigma} - \frac{1}{\sigma^2}\cdot\frac{1}{2}\sum_{i=1}^{n}(y^{(i)}-\theta^T x^{(i)})^2$$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ

$$= -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y^{(i)}-\theta^T x^{(i)})^2$$

$$= -\frac{1}{2}\sum_{i=1}^{n}(y^{(i)}-\theta^T x^{(i)})^2$$

3. Solve

$$\hat{\theta} = \underset{\theta}{\operatorname{argmax}}\, L(\theta)$$

$$= \text{argmax}\ \left(-\frac{1}{2}\sum_{i=1}^{n}(y^{(i)}-\theta^T x^{(i)})^2\right)$$

$$= \text{argmin}\ \left(\frac{1}{2}\sum_{i=1}^{n}(y^{(i)}-\theta^T x^{(i)})^2\right)$$

With probabilistic assumptions on the data, least-squares regression corresponds to finding the MLE of θ

•MLE – Logistic Regression

$$y_i \mid x_i \sim \text{Bern}(\sigma(\mathbf{w}^\top x_i))$$

1. Determine formula for $LL(\theta)$

$$LL(\theta) = \log \prod_{i=1}^{n} P_\theta\left(y^{(i)} | x^{(i)}\right) = \sum_{i=1}^{n} \log P_\theta\left(y^{(i)} | x^{(i)}\right)$$

$$p(y \mid x; \theta) = (h_\theta(x))^y \, (1 - h_\theta(x))^{1-y}$$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ

3. Solve

Mileage (in kmpl)	*Car Price (in cr)*
9.8	High
9.12	Low
9.5	High
10	Low

Bernoulli MLE Estimation

$X_1, X_2, \ldots, X_n$ where $X_i \sim \text{Ber}(p)$. PMF of a Bernoulli $p^{X_i}(1-p)^{1-X_i}$

$$P_\theta(Y = 1 | X = x) = h_\theta(x) = \frac{1}{1 + e^{-\theta^\top x}}$$

$$P_\theta(Y = 0 | X = x) = 1 - h_\theta(x) = \frac{e^{-\theta^\top x}}{1 + e^{-\theta^\top x}}$$

MLE answers the question: For which parameter value does the observed data have the largest probability?

•MLE – Logistic Regression

$$y_i \mid x_i \sim \text{Bern}(\sigma(\mathbf{w}^\intercal x_i))$$

1. Determine formula for $LL(\theta)$

$$L(\theta) = \prod_{i=1} P(Y = y^{(i)} | X = \mathbf{x}^{(i)})$$
$$= \prod_{i=1}^{n} \sigma(\theta^T \mathbf{x}^{(i)})^{y^{(i)}} \cdot \left[1 - \sigma(\theta^T \mathbf{x}^{(i)})\right]^{(1-y^{(i)})}$$

$$LL(\theta) = \sum_{i=1}^{n} y^{(i)} \log \sigma(\theta^T \mathbf{x}^{(i)}) + (1 - y^{(i)}) \log[1 - \sigma(\theta^T \mathbf{x}^{(i)})]$$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ

$$\frac{\partial LL(\theta)}{\partial \theta_j} = \frac{\partial}{\partial \theta_j} y \log \sigma(\theta^T \mathbf{x}) + \frac{\partial}{\partial \theta_j} (1 - y) \log[1 - \sigma(\theta^T \mathbf{x}]$$
$$= \left[\frac{y}{\sigma(\theta^T \mathbf{x})} - \frac{1 - y}{1 - \sigma(\theta^T \mathbf{x})}\right] \frac{\partial}{\partial \theta_j} \sigma(\theta^T \mathbf{x})$$
$$= \left[\frac{y}{\sigma(\theta^T \mathbf{x})} - \frac{1 - y}{1 - \sigma(\theta^T \mathbf{x})}\right] \sigma(\theta^T \mathbf{x})[1 - \sigma(\theta^T \mathbf{x})]\mathbf{x}_j$$
$$= \left[\frac{y - \sigma(\theta^T \mathbf{x})}{\sigma(\theta^T \mathbf{x})[1 - \sigma(\theta^T \mathbf{x})]}\right] \sigma(\theta^T \mathbf{x})[1 - \sigma(\theta^T \mathbf{x})]\mathbf{x}_j$$
$$= [y - \sigma(\theta^T \mathbf{x})]\mathbf{x}_j$$

3. Solve

$$\theta := \theta + \alpha \nabla_\theta \ell(\theta)$$

MLE answers the question: For which parameter value does the observed data have the largest probability?

•MLE – Logistic Regression

$$y_i \mid x_i \sim \text{Bern}(\sigma(\mathbf{w}^\intercal x_i))$$

1. Determine formula for $LL(\theta)$

$$L(\theta) = \prod_{i=1} P(Y = y^{(i)} | X = \mathbf{x}^{(i)})$$
$$= \prod_{i=1}^{n} \sigma(\theta^T \mathbf{x}^{(i)})^{y^{(i)}} \cdot \left[1 - \sigma(\theta^T \mathbf{x}^{(i)})\right]^{(1-y^{(i)})}$$

$$LL(\theta) = \sum_{i=1}^{n} y^{(i)} \log \sigma(\theta^T \mathbf{x}^{(i)}) + (1 - y^{(i)}) \log[1 - \sigma(\theta^T \mathbf{x}^{(i)})]$$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ

$$\frac{\partial LL(\theta)}{\partial \theta_j} = \sum_{i=1}^{n} \left[y^{(i)} - \sigma(\theta^T \mathbf{x}^{(i)})\right] x_j^{(i)}$$

3. Solve

MLE answers the question: For which parameter value does the observed data have the biggest probability?

• MLE-Summary

- Consider a sample of n i.i.d. random variables $X_1, X_2, \ldots, X_n$, drawn from a distribution $f(X_i|\theta)$

- θ_{MLE} maximizes the likelihood of data, $L(\theta)$ and $LL(\theta)$

$$L(\theta) = \prod_{i=1}^{n} f(X_i|\theta) \qquad \hat{\theta} = \underset{\theta}{\operatorname{argmax}}\, L(\theta)$$

$$LL(\theta) = \log L(\theta) = \log \prod_{i=1}^{n} f(X_i|\theta) = \sum_{i=1}^{n} \log f(X_i|\theta)$$

$$\theta_{MLE} = \arg\max_{\theta} LL(\theta)$$

Additional Problems in MLE

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•MLE – Discrete - PMF

Example 1: Suppose that X is a discrete random variable with the following probability mass function: where $0 \leq \theta \leq 1$ is a parameter. The following 10 independent observations

X	0	1	2	3
$P(X)$	$2\theta/3$	$\theta/3$	$2(1-\theta)/3$	$(1-\theta)/3$

were taken from such a distribution: (3,0,2,1,3,2,1,0,2,1). What is the maximum likelihood estimate of θ.

1. Determine formula for $LL(\theta)$
2. Differentiate $LL(\theta)$ w.r.t. (each) θ
3. Solve

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•MLE – Discrete - PMF

1. Determine formula for $LL(\theta)$

   L(ɵ) = P(X=3)*P(X=0)*P(X=2)*……..*P(X=1)

   $= ((1-\theta) / 3)^2 * (2\theta / 3)^2 * (2(1-\theta) / 3)^3 * (\theta / 3)^3$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ

3. Solve

X	P(X)
3	(1-ɵ) / 3
0	2ɵ / 3
2	2(1-ɵ) / 3
1	ɵ/ 3
3	(1-ɵ) / 3
2	2(1-ɵ) / 3
1	ɵ/ 3
0	2ɵ / 3
2	2(1-ɵ) / 3
1	ɵ/ 3

•MLE – Discrete - PMF

1. Determine formula for $LL(\theta)$

$L(\theta) = P(X=3)*P(X=0)*P(X=2)*\ldots\ldots*P(X=1)$

$= ((1-\theta) / 3)^2 * (2\theta / 3)^2 * (2(1-\theta) / 3)^3 * (\theta / 3)^3$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ

$LL(\theta) = \log [((1-\theta) / 3)^2 * (2\theta / 3)^2 * (2(1-\theta) / 3)^3 * (\theta / 3)^3]$

$= \log ((1-\theta) / 3)^2 + \log (2\theta / 3)^2 + \log (2(1-\theta) / 3)^3 * \log(\theta / 3)^3$

$= 2\log ((1-\theta) / 3) + 2\log (2\theta / 3) + 3 \log (2(1-\theta) / 3) + 3\log(\theta / 3)$

$=2(\log ((1-\theta) - \log 3) + 2(\log (2\theta) - \log 3) + 3(\log (2(1-\theta)) - \log 3) + 3(\log(\theta) - \log 3)$

Gradient ($LL(\theta)$) $= -\frac{2}{(1-\theta)} + \frac{2}{\theta} - \frac{3}{(1-\theta)} + \frac{3}{\theta}$

3. Solve

$$= \frac{-5\theta + 5 - 5\theta}{(1-\theta)\theta} = \frac{-10\theta + 5}{(1-\theta)\theta} = 0 \rightarrow \theta = 0.5$$

•Additional Exercise for Student's Practice

Consider inputs x_i which are real valued attributes and the outputs y_i which are real valued of the form $y_i = f(x_i) + e_i$, where $f(x_i)$ is the true function and e_i is a random variable representing laplacian noise with PDF given by

$$f(y_i/\theta) = \frac{1}{2\theta} * e^{\frac{-|y_i - \mu|}{\theta}}$$

Implementing a linear regression model of the form , $h(x_i) = \sum_{i=0}^{n} \theta_i x_i$ and $\mu = h(x_i)$, find the maximum likelihood estimator of è. Comment on the loss function.

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•Additional Exercise for Student's Practice

1. Determine formula for $LL(\theta)$

$$L(\theta) = L(\theta) = \frac{1}{2\theta} e^{\frac{-|\mathbf{y}_1 - \mu|}{\theta}} * \frac{1}{2\theta} e^{\frac{-|\mathbf{y}_2 - \mu|}{\theta}} * \frac{1}{2\theta} e^{\frac{-|\mathbf{y}_3 - \mu|}{\theta}} * \ldots\ldots * \frac{1}{2\theta} e^{\frac{-|\mathbf{y}_n - \mu|}{\theta}}$$

$$= \left(\prod_{i=1}^{n} \frac{1}{2\theta} e^{\frac{-|\mathbf{y}_i - \mu|}{\theta}}\right)$$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ

$$LL(\theta) = \ln\left(\prod_{i=1}^{n} \frac{1}{2\theta} e^{\frac{-|\mathbf{y}_i - \mu|}{\theta}}\right) = -\ln(2\theta) \sum_{i=1}^{n} 1 + \sum_{i=1}^{n} \ln\left(e^{\frac{-|\mathbf{y}_i - \mu|}{\theta}}\right)$$

$$= \underset{\theta}{argmax} - n \ln(2\theta) - \sum_{i=1}^{n} \frac{|\mathbf{y}_i - \mu|}{\theta}$$

$$= \underset{\theta}{argmin}\ n \ln(2\theta) + \sum_{i=1}^{n} \frac{|\mathbf{y}_i - \mu|}{\theta}$$

3. Solve

$$\text{Gradient}\ (LL(\theta)) = \frac{n}{(\theta)} - \frac{\sum_{i=1}^{n} |\mathbf{y}_i - \mu|}{(\theta^2)} = 0$$

$$\frac{n}{(\theta)} = \frac{\sum_{i=1}^{n} |\mathbf{y}_i - \mu|}{(\theta^2)}$$

$$\theta = \frac{\sum_{i=1}^{n} |\mathbf{y}_i - \mu|}{n}$$

Instead of MSE, MAE is the maximum likelihood hypothesis. So MAE is appropriate for the loss function

•Maximum A Posteriori (MAP) Analysis

1. Determine prior probability

$$\theta_{MAP} = \arg\max f(\theta | X_1, X_2, \dots, X_n)$$

2. Find the posterior probability for every distinct prior

Brute Force MAP Hypothesis

$$h_{MAP} = \underset{h \in H}{argmax}\ P(h|D)$$

$$= \underset{h \in H}{argmax}\ \frac{P(D|h)P(h)}{P(D)}$$

$$= \underset{h \in H}{argmax}\ P(D|h)P(h)$$

3. Choose the posterior with highest h_{MAP} value

Maximum a Posteriori (MAP) Estimator of θ is the value of θ that maximizes the posterior distribution of θ.

Best hypothesis ≈ most probable hypothesis

•Maximum A Posteriori Estimation (MAP)

1. Find the prior probability

2. Derive the posterior probability

3. Differentiate posterior w.r.t. (each) θ

4. Solve

Maximum a Posteriori (MAP) Estimator of θ is the value of θ that maximizes the posterior distribution of θ.

Best hypothesis ≈ most probable hypothesis

Example

1. Example on MAP algorithm:

Let X be continuous random variable with probability density function P(X) given by:

$$f(x) = \begin{cases} 2x, & 0 \le x \le 1 \\ 0, & otherwise \end{cases}$$

Given another distribution $p(Y|X = x) = x(1-x)^{y-1}$ Find MAP estimate of X given Y=3

h_{MAP} = P(X | Y=3)

= P(Y=3 | X) * P(X)

= $x(1-x)^{y-1} * 2x$

To find the parameter x , differentiate the function & equate to zero.

$$\frac{d(P(X|Y=3)}{dx} = 0$$

$$\frac{d(x(1-x)2 * 2x)}{dx} = 0$$

$$\frac{d(2x^2-4x^3+2x^4)}{dx} = 0$$

$$4x-12x^2+8x^3 = 0$$

$x = \{0, 0.5, 1\}$

Example :

1. Example on MAP algorithm:

Let X be continuous random variable with probability density function P(X) given by:

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Given another distribution $p(Y|X = x) = x(1-x)^{y-1}$ Find MAP estimate of X given Y=3

h_{MAP} = P(X | Y=3)

= P(Y=3 | X) * P(X)

= $x(1-x)^{y-1} * 2x$

x = {0, 0.5, 1}

P(X|Y=3) = {0, 0.125 ,0}

Most Probable Classification of New Instances

- So far we've sought the most probable hypothesis given the data D (i.e., h_{MAP})
- **Given new instance x, what is its most probable classification?**
  - $h_{MAP}(x)$ is not the most probable classification!
  - What's most probable classification of x?

Consider:

Three possible hypotheses:

$P(h_1|D) = .4$, $P(h_2|D) = .3$, $P(h_3|D) = .3$

Given new instance x, classification given by above 3 hypotheses is

$h_1(x) = +$, $h_2(x) = -$, $h_3(x) = -$

$P(\oplus|h_1) = 1$ and $P(\ominus|h_1) = 0$

May be the classification for X is **+**

ML setting

- Bayesian Analysis
  - start with some belief about the system, called a prior.
  - Then we obtain some data and use it to update our belief.
  - The outcome is called a posterior.
  - Should we obtain even more data, the old posterior becomes a new prior and the cycle repeats.
  - P(h | D) a posterior determines the class label
  - MLE and MAP are the same if the prior is uniform
  - This forms the basis for Naïve Bayes classifier → More on this will be discussed in later part of this module

• Bayes' Optimal Classifier

- The most probable classification of the new instance is obtained by combining **the predictions of all hypotheses, weighted by their posterior probabilities.**
- v_j from some set V, then the probability $P(v_j \mid D)$ that the correct classification for the new instance is v_j $P(v_j|D) = \sum_{h_i \in H} P(v_j|h_i)P(h_i|D)$
- The optimal classification of the new instance is the value vj for which $P(v_j \mid D)$ is maximum

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Example 1 : Bayes Optimal Classifier

Bayes optimal classification:

$$\arg\max_{v_j \in V} \sum_{h_i \in H} P(v_j|h_i)P(h_i|D)$$

- Example:

$P(h_1|D) = .4$, $P(-|h_1) = 0$, $P(+|h_1) = 1$

$P(h_2|D) = .3$, $P(-|h_2) = 1$, $P(+|h_2) = 0$

$P(h_3|D) = .3$, $P(-|h_3) = 1$, $P(+|h_3) = 0$

therefore

$$\sum_{h_i \in H} P(+|h_i)P(h_i|D) = .4$$

$$\sum_{h_i \in H} P(-|h_i)P(h_i|D) = .6$$

and

$$\boxed{\arg\max_{v_j \in V} \sum_{h_i \in H} P(v_j|h_i)P(h_i|D) = -}$$

Gibbs Classifier

- Bayes optimal classifier provides best result, but can be expensive if many hypotheses.
- Gibbs algorithm:
  - Choose one hypothesis at random, according to posterior prob. Distribution over h , $P(h|D)$
  - Use this h to classify new instance
- Surprising fact: under certain conditions, the expected misclassification error for the Gibbs algorithm is at most twice the expected error of the Bayes optimal classifier

$$E[error_{Gibbs}] \leq 2E[error_{BayesOptional}]$$

- Suppose correct, uniform prior distribution over *H*, then
  - Pick any hypothesis from *Version space*, with uniform probability
  - Its expected error no worse than twice Bayes optimal

Parameter Estimation in ML - Summary

Assumption: Data are__

- IID samples: $X_1 \ldots X_n$
  where all X_i are independent and have the same distribution.
- Either same PMF (discrete) or same PDF (continuous)
- $f(X \mid \theta)$
  Likelihood of different values of X depends on the values of our parameters θ
  f( ) is either PDF or PMF

$$\text{Parameters}: (\theta_0, \theta_1, \theta_2)$$

Find $\boldsymbol{\theta} = (\boldsymbol{\theta_0}, \boldsymbol{\theta_1}, \boldsymbol{\theta_2})$ from the data $\mathbf{X_i}$

Maximum Likelihood Estimation (MLE)

select that parameters θ that make the observed data the most likely

$$f(X_1, X_2 \ldots X_n \mid \boldsymbol{\theta})$$

Maximum A Posteriori (MAP)

choose the parameters θ that is the most likely, given the data

$$f(\boldsymbol{\theta} \mid X_1, X_2 \ldots X_n)$$

If the sample is large, MLE will yield an excellent estimator of θ

When no prior information is available, all hypothesis are equally likely i.e. p(hi) = p(hj)

Naïve Bayes Classifier

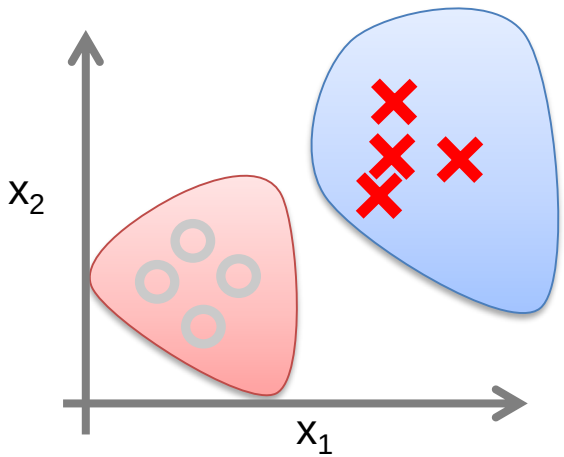
Work Integrated Learning Programmes

Refresh from Pre-Mid Term

Decision Theory: Interpretation

Model Building

Generative



Likelihood

Class Prior Probability

$$P(c \mid x) = \frac{P(x \mid c)P(c)}{P(x)}$$

Posterior Probability

Predictor Prior Probability

$$P(c \mid \mathrm{X}) = P(x_1 \mid c) \times P(x_2 \mid c) \times \cdots \times P(x_n \mid c) \times P(c)$$

Sky	AirTemp	Humidity	Wind	Forecast	Enjoy Sport?
Sunny	Warm	Normal	Strong	Same	Yes
Sunny	Warm	High	Strong	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	Normal	Breeze	Same	Yes
Sunny	Hot	Normal	Breeze	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	High	Strong	Change	Yes
Rainy	Warm	Normal	Breeze	Same	Yes

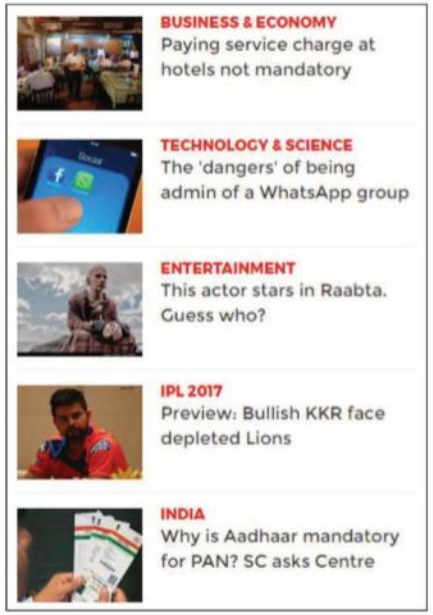
$$P(Y \mid X_1 X_2 \ldots X_n) = \frac{P(X_1 X_2 \ldots X_d \mid Y)P(Y)}{P(X_1 X_2 \ldots X_d)}$$

Known as generative models, because by sampling from them it is possible to generate synthetic data points in the input space.

Eg., Gaussians, **Naïve Bayes**, Mixtures of multinomials **, Mixtures of Gaussians**, Bayesian networks

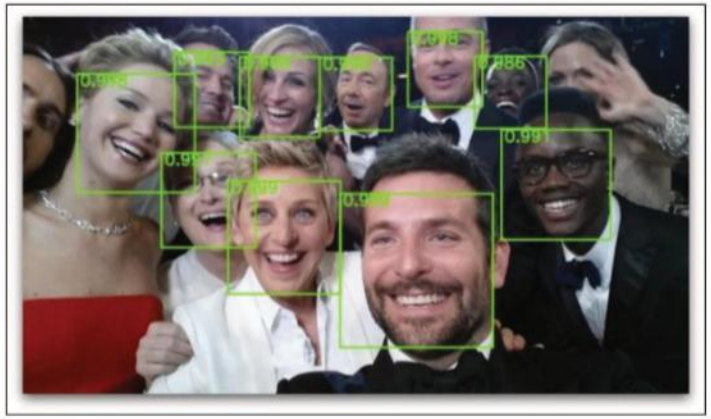
Naïve Bayes Classifier - Applications

Categorizing News



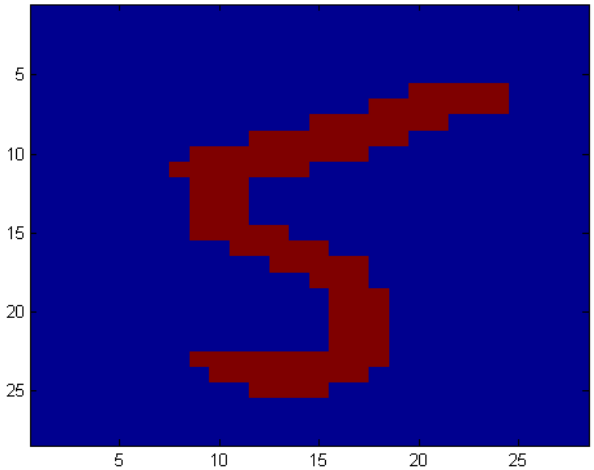
Email Spam Detection

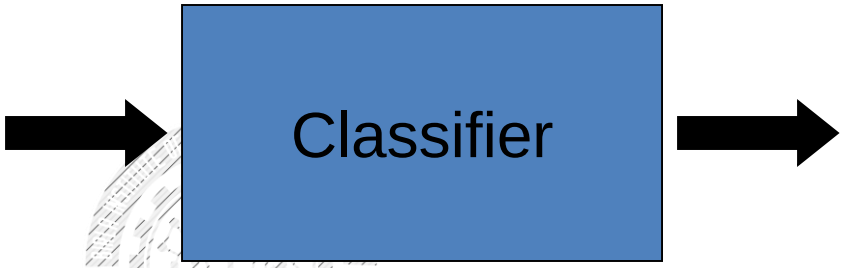
Face Recognition



Sentiment Analysis

Example: Digit Recognition





- $X_1,\ldots,X_n \in \{0,1\}$ (Black vs. White pixels)
- $Y \in \{5,6\}$ (predict whether a digit is a 5 or a 6)

Generative models for classification

$$p(y|x) = \frac{p(x|y)p(y)}{p(x)}.$$

- For binary classification the denominator is given by

  p(x) = p(x|y = 1)p(y = 1) + p(x|y = 0)p(y = 0)

- if we're calculating p(y|x) in order to make a prediction, then we don't actually need to calculate the denominator, since

$$\begin{aligned} \arg\max_y p(y|x) &= \arg\max_y \frac{p(x|y)p(y)}{p(x)} \\ &= \arg\max_y p(x|y)p(y). \end{aligned}$$

The Bayes Classifier

$$P(Y=5|X_1,\ldots,X_n) = \frac{P(X_1,\ldots,X_n|Y=5)P(Y=5)}{P(X_1,\ldots,X_n|Y=5)P(Y=5)+P(X_1,\ldots,X_n|Y=6)P(Y=6)}$$

$$P(Y=6|X_1,\ldots,X_n) = \frac{P(X_1,\ldots,X_n|Y=6)P(Y=6)}{P(X_1,\ldots,X_n|Y=5)P(Y=5)+P(X_1,\ldots,X_n|Y=6)P(Y=6)}$$

- To classify, we'll simply compute these two probabilities and predict based on which one is greater

Naïve Bayes conditional Independence assumption

- Naïve Bayes assumes X_i are conditionally independent given Y

$$P(X_1|X_2,Y) = P(X_1|Y)$$

- **Assumption**:

$$P(X_1,\cdots,X_n|Y) = \prod_{j=1}^{n} P(X_j|Y)$$

i.e., X_i and X_j are <u>conditionally independent</u> given Y for $i \neq j$

Slide credit: Tom Mitchell

Naïve Bayes classifier: Prediction

Goal of learning P(Y|X) where X = <X_1,…, X_n>

- Bayes rule:

$$P(Y = y_k | X_1, \cdots, X_n) = \frac{P(Y = y_k) P(X_1, \cdots, X_n | Y = y_k)}{\sum_j P(Y = y_j) P(X_1, \cdots, X_n | Y = y_j)}$$

- Assume conditional independence among X_i's:

$$P(Y = y_k | X_1, \cdots, X_n) = \frac{P(Y = y_k) \Pi_i P(X_i | Y = y_k)}{\sum_j P(Y = y_j) \Pi_i P(X_i | Y = y_j)}$$

- Classify New Instance(*x*) : Pick the most probable (MAP) Y for

$$X_{new} = < X_1, \ldots, X_n >$$

$$\hat{Y} \leftarrow \underset{y_k}{\operatorname{argmax}} P(Y = y_k) \Pi_i P(X_i | Y = y_k)$$

Prior **Likelihood**

Slide credit: Tom Mitchell

- Example: Play Tennis

Likelihood

Class Prior Probability

$$P(c \mid x) = \frac{P(x \mid c)P(c)}{P(x)}$$

Posterior Probability

Predictor Prior Probability

$$P(c \mid X) = P(x_1 \mid c) \times P(x_2 \mid c) \times \cdots \times P(x_n \mid c) \times P(c)$$

$$\hat{Y} \leftarrow \underset{y_k}{\operatorname{argmax}}\, P(Y = y_k)\Pi_i P(X_i \mid Y = y_k)$$

P(X|Y)~ Multinom(π, n) → Multinomial NB (X_i – multinomial)

P(Y) ~ Ber(p

Sky	AirTemp	Humidity	Wind	Forecast	Enjoy Sport?
Sunny	Warm	Normal	Strong	Same	Yes
Sunny	Warm	High	Strong	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	Normal	Breeze	Same	Yes
Sunny	Hot	Normal	Breeze	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	High	Strong	Change	Yes
Rainy	Warm	Normal	Breeze	Same	Yes

- Example: Play Tennis – Learning Phase

Look up tables

Maximum likelihood estimates (MLE's):

$$\hat{\theta}_{ijk} = \hat{P}(X_i = x_{ij} | Y = y_k) = \frac{\#D\{X_i = x_{ij} \wedge Y = y_k\}}{\#D\{Y = y_k\}} \qquad \hat{\pi}_k = \hat{P}(Y = y_k) = \frac{\#D\{Y = y_k\}}{|D|}$$

Number of items in dataset D for which $Y=y_k$

Sky	Play=Yes	Play=No
Sunny	3/3	2/4
Rainy	0/3	2/4

Wind	Play=Yes	Play=No
Strong	2/3	3/4
Breeze	1/3	1/4

AirTemp	Play=Yes	Play=No
Hot	0/3	1/4
Warm	3/3	1/4
Cold	0/3	2/4

Forecast	Play=Yes	Play=No
Same	2/3	2/4
Change	1/3	2/4

Humidity	Play=Yes	Play=No
High	1/3	3/4
Normal	2/3	1/4

Sky	AirTemp	Humidity	Wind	Forecast	Enjoy Sport?
Sunny	Warm	Normal	Strong	Same	Yes
Sunny	Warm	High	Strong	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	Normal	Breeze	Same	Yes
Sunny	Hot	Normal	Breeze	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	High	Strong	Change	Yes
Rainy	Warm	Normal	Breeze	Same	Yes

- Example: Play Tennis - Testing Phase

MAP rule

$$Y^{new} \leftarrow \arg\max_{y_k} \; P(Y = y_k) \prod_i P(X_i^{new} | Y = y_k)$$

$$Y^{new} \leftarrow \arg\max_{y_k} \; \pi_k \prod_i \theta_{ijk}$$

•P(Enjoy=Yes | X) = P(X | Enjoy=Yes). P(Enjoy=Yes) / P(X)

•= P(X | Enjoy=Yes). P(Enjoy=Yes)

•= P(X | Enjoy=Yes). (3/7)

•= P(Sunny | Enjoy=Yes). P(Warm | Enjoy=Yes). P(Normal | Enjoy=Yes). P(Strong | Enjoy=Yes). P(Change | Enjoy=Yes).(3/7)

•= (3/3) . (3/3) . (2/3) . (2/3) . (1/3) . (3/7)

•=0.0635

$$\hat{\pi}_k = \hat{P}(Y = y_k) = \frac{\#D\{Y = y_k\}}{|D|}$$

Sky	AirTemp	Humidity	Wind	Forecast	Enjoy Sport?
Sunny	Warm	Normal	Strong	Same	Yes
Sunny	Warm	High	Strong	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	Normal	Breeze	Same	Yes
Sunny	Hot	Normal	Breeze	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	High	Strong	Change	Yes
Sunny	Warm	Normal	Strong	Change	????
Rainy	Warm	Normal	Breeze	Same	????

•P(Enjoy=Yes | X) > P(Enjoy=No | X) → EnjoySport = **Yes**

•P(Enjoy=No | X) = P(X | Enjoy=No). P(Enjoy=No) / P(X)

•= P(X | Enjoy=No). P(Enjoy=No)

•= P(X | Enjoy=No). (4/7)

•= P(Sunny | Enjoy=No). P(Warm | Enjoy=No). P(Normal | Enjoy=No). P(Strong | Enjoy=No). P(Change | Enjoy=No). (4/7)

•= (2/4) . (1/4) . (1/4) . (3/4) . (2/4) . (4/7)

- Train Naïve Bayes (examples) for each* value y_k

  estimate $\pi_k \equiv P(Y = y_k)$

  for each* value x_{ij} of each attribute X_i

  estimate $\theta_{ijk} \equiv P(X_i = x_{ij} | Y = y_k)$

- Classify (X^{new})

$$Y^{new} \leftarrow \arg\max_{y_k} \; P(Y = y_k) \prod_i P(X_i^{new} | Y = y_k)$$

$$Y^{new} \leftarrow \arg\max_{y_k} \; \pi_k \prod_i \theta_{ijk}$$

- Example: Play Tennis

•*P(Enjoy=Yes | X) = P(X | Enjoy=Yes). P(Enjoy=Yes) / P(X)*

•*= P(X | Enjoy=Yes). P(Enjoy=Yes)*

•*= P(X | Enjoy=Yes). (3/7)*

•*= P(Rainy | Enjoy=Yes). P(Warm | Enjoy=Yes). P(Normal | Enjoy=Yes). P(Breeze | Enjoy=Yes) P(Same | Enjoy=Yes).(3/7)*

•*= (0/3) . (3/3) . (2/3) . (1/3) . (2/3) . (3/7)*

•

Sky	Enjoy Sport?
Sunny	Yes
Sunny	No
Rainy	No
Sunny	Yes
Sunny	No
Rainy	No
Sunny	Yes
Rainy	????

AirTemp	Humidity	Wind	Forecast	Enjoy Sport?
Warm	Normal	Strong	Same	Yes
Warm	High	Strong	Same	No
Cold	High	Strong	Change	No
Warm	Normal	Breeze	Same	Yes
Hot	Normal	Breeze	Same	No
Cold	High	Strong	Change	No
Warm	High	Strong	Change	Yes
Warm	Normal	Breeze	Same	????

•*P(Enjoy=No | X) = P(X | Enjoy=No). P(Enjoy=No) / P(X)*

•*= P(X | Enjoy=No). P(Enjoy=No)*

•*= P(X | Enjoy=No). (4/7)*

•*= P(Rainy | Enjoy=No). P(Warm | Enjoy=No). P(Normal | Enjoy=No). P(Breeze | Enjoy=No). P(Same | Enjoy=No). (4/7)*

Laplace Smoothing

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Smoothing

If one of the conditional probabilities is zero, then the entire expression becomes zero

- Technique for smoothing categorical data.
- A small-sample correction, or **pseudo-count**, will be incorporated in every probability estimate.
- No probability will be zero.

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Smoothing

Probability estimation:

$$\text{Original}: P(A_i \mid C) = \frac{N_{ic}}{N_c}$$

$$\text{Laplace}: P(A_i \mid C) = \frac{N_{ic} + 1}{N_c + d}$$

$$\text{m-estimate}: P(A_i \mid C) = \frac{N_{ic} + mp}{N_c + m}$$

Bayesian approach

d: number of distinct values of an attribute

N_c: number of instances in the class

N_{ic}: number of instances having attribute value A_i in class *c*

p: prior probability of the class

m: constant called the **equivalent sample size**, which determines how heavily to weight p relative to the observed data

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- Example: Play Tennis

•*P(Enjoy=Yes | X) = P(X | Enjoy=Yes). P(Enjoy=Yes) / P(X)*

•*= P(X | Enjoy=Yes). P(Enjoy=Yes)*

•*= P(X | Enjoy=Yes). (3/7)*

•*= P(Rainy | Enjoy=Yes). P(Warm | Enjoy=Yes). P(Normal | Enjoy=Yes). P(Breeze | Enjoy=Yes). P(Same | Enjoy=Yes).(3/7)*

•*= (0+1/3+2) . (3 +1/3+3) . (2+1/3+2) . (1+1/3+2) . (2+1/3+2) . (3/7)*

$$\hat{\theta}_i = \frac{x_i + \alpha}{N + \alpha d} \qquad (i = 1, \ldots, d),$$

•

- ***For Laplace smoothing α = 1***

AirTemp	Humidity	Wind	Forecast	Enjoy Sport?
Warm	Normal	Strong	Same	Yes
Warm	High	Strong	Same	No
Cold	High	Strong	Change	No
Warm	Normal	Breeze	Same	Yes
Hot	Normal	Breeze	Same	No
Cold	High	Strong	Change	No
Warm	High	Strong	Change	Yes
Warm	Normal	Breeze	Same	????

•*P(Enjoy=No | X) = P(X | Enjoy=No). P(Enjoy=No) / P(X)*

•*= P(X | Enjoy=No). P(Enjoy=No)*

•*= P(X | Enjoy=No). (4/7)*

•*= P(Rainy | Enjoy=No). P(Warm | Enjoy=No). P(Normal | Enjoy=No). P(Breeze | Enjoy=No). P(Same | Enjoy=No). (4/7)*

- Example: Play Tennis

•*P(Enjoy=Yes | X) = P(X | Enjoy=Yes). P(Enjoy=Yes) / P(X)*

•*= P(X | Enjoy=Yes). P(Enjoy=Yes)*

•*= P(X | Enjoy=Yes). (3/7)*

•*= P(Rainy | Enjoy=Yes). P(Warm | Enjoy=Yes). P(Normal | Enjoy=Yes). P(Breeze | Enjoy=Yes). P(Same | Enjoy=Yes).(3/7)*

•*= (1/5) . (4/6) . (3/5) . (2/5) . (3/5) . (3/7)*

•*=0.00822*

Sky	AirTemp	Humidity	Wind	Forecast	Enjoy Sport?
Sunny	Warm	Normal	Strong	Same	Yes
Sunny	Warm	High	Strong	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	Normal	Breeze	Same	Yes
Sunny	Hot	Normal	Breeze	Same	No
Rainy	Cold	High	Strong	Change	No
Sunny	Warm	High	Strong	Change	Yes
Sunny	Warm	Normal	Strong	Change	????
Rainy	Warm	Normal	Breeze	Same	????

•*P(Enjoy=Yes | X) > P(Enjoy=No | X)* → EnjoySport = **Yes**

•*P(Enjoy=No | X) = P(X | Enjoy=No). P(Enjoy=No) / P(X)*

•*= P(X | Enjoy=No). P(Enjoy=No)*

•*= P(X | Enjoy=No). (4/7)*

•*= P(Rainy | Enjoy=No). P(Warm | Enjoy=No). P(Normal | Enjoy=No). P(Breeze | Enjoy=No). P(Same | Enjoy=No). (4/7)*

•*= (3/6) . (2/7) . (2/6) . (2/6) . (3/6) . (4/7)*

Naïve Bayes: Continuous Features

- X_i can be continuous

Naïve Bayes classifier:

$$Y = \arg\max_{y} P(Y = y) \prod_{i} P(X_i | Y = y)$$

Assumption: $P(X_i|Y)$ has a **Gaussian** distribution

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The Gaussian Probability Distribution

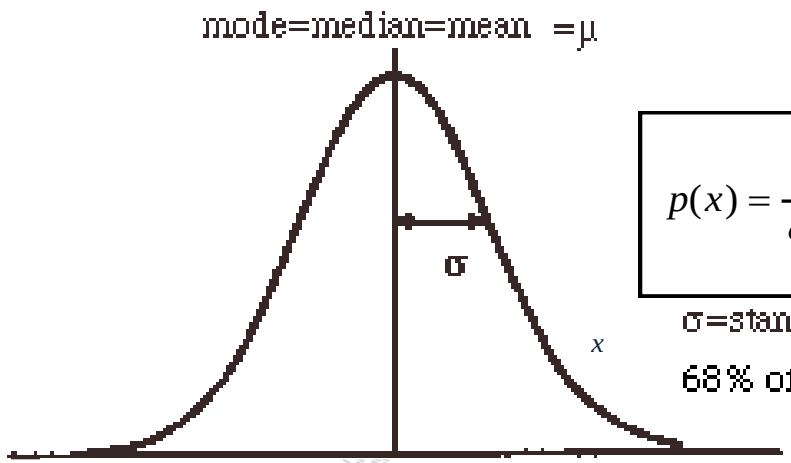
- It is a continuous distribution with pdf:

μ = mean of distribution

σ^2 = variance of distribution

x is a continuous variable $(-\infty \leq x \leq \infty)$

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad gaussian$$

σ=standard deviation

68% of area within ±1σ

Continuous Features : learning and prediction

- For each target value Y_k (MLE estimate)

$P(Y = y_k) \leftarrow$ No. of instances with Y_k class/No. of Total instances

- For each attribute value X_i estimate $P(X_i|Y = y_k)$
  - class conditional mean , variance

- Classify New Instance(x)

Pick the most probable (MAP) Y

$$\hat{Y} \leftarrow \underset{y_k}{\operatorname{argmax}}\, P(Y = y_k)\Pi_i P(X_i|Y = y_k)$$

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Continuous Features : learning

- $P(X_i|Y)$ is Gaussian
- Training: estimate mean and standard deviation
  - $\mu_i = E[X_i|Y = y]$
  - $\sigma_i^2 = E[(X_i - \mu_i)^2|Y = y]$

X_1	X_2	X_3	Y
2	3	1	1
-1.2	2	0.4	1
1.2	0.3	0	0
2.2	1.1	0	1

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Continuous Features : learning

X_1	X_2	X_3	Y
2	3	1	1
-1.2	2	0.4	1
1.2	0.3	0	0
2.2	1.1	0	1

- $\mu_i = E[X_i|Y = y]$
- $\sigma_i^2 = E[(X_i - \mu_i)^2|Y = y]$

- $\mu_1 = E[X_1|Y = 1] = \frac{2+(-1.2)+2.2}{3} = 1$
- $\sigma_1^2 = E[(X_1 - \mu_1)|Y = 1] =$

$$\frac{(2-1)^2 + (-1.2-1)^2 + (2.2-1)^2}{3} = 2.43$$

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- Example: Play Tennis

$P(X|Y) \sim N(\mu, \sigma^2) \rightarrow$ GaussianNB (X_i – real valued)

•*P(Enjoy=Yes | X) = P(X | Enjoy=Yes). P(Enjoy=Yes) / P(X)*

•*= P(X | Enjoy=Yes). P(Enjoy=Yes)*

•*= P(X | Enjoy=Yes). (3/7)*

•*= P(Rainy | Enjoy=Yes). P(Warm | Enjoy=Yes). P(60 | Enjoy=Yes). P(Breeze | Enjoy=Yes). P(Same | Enjoy=Yes).(3/7)*

•*= (1/3) . (3/3) . 0.15*10^{-95} . (1/3) . (2/3) . (3/7)*

$$\mu_i = E[X_i|Y = yes] = 84.33$$
$$\sigma_i^2 = E[(X_i - \mu_i)^2|Y = yes] = 1.15$$

•
$$\mu_i = E[X_i|Y = no] = 72.5$$
$$\sigma_i^2 = E[(X_i - \mu_i)^2|Y = no] = 17.08$$

Humidity	Enjoy Sport?
85	Yes
80	No
70	No
83	Yes
90	No
50	No
85	Yes
60	????

AirTemp	Sky	Wind	Forecast	Enjoy Sport?
Warm	Sunny	Strong	Same	Yes
Warm	Sunny	Strong	Same	No
Cold	Rainy	Strong	Change	No
Warm	Rainy	Breeze	Same	Yes
Hot	Sunny	Breeze	Same	No
Cold	Rainy	Strong	Change	No
Warm	Sunny	Strong	Change	Yes
Warm	Rainy	Breeze	Same	????

•*P(Enjoy=No | X) = P(X | Enjoy=No). P(Enjoy=No) / P(X)*

•*= P(X | Enjoy=No). P(Enjoy=No)*

•*= P(X | Enjoy=No). (4/7)*

•*= P(Rainy | Enjoy=No). P(Warm | Enjoy=No). P(60 | Enjoy=No). P(Breeze | Enjoy=No). P(Same | Enjoy=No). (4/7)*

• Additional Exercises – For Student's Practice

As a part of efforts to improve students' performance in the exams, you have been given the data showing number of study hours spent by students, their gender and their final results as pass or fail. Using this sample dataset, apply Naïve Bayes classification technique, to classify the below test case:

{No of study hours = 3.5, Gender="male"} either as "Pass", or "Fail".

Tip: $P(X|Y) \sim N(\mu, \sigma^2) \rightarrow$ GaussianNB (X_i – real valued)

No of study hours	Gender	Final result
4.5	Male	Pass
7	Female	Pass
2	Male	Fail
4	Female	Fail
2.5	Male	Fail
3	Female	Fail
8.3	Male	Fail
8	Female	Pass
9	Male	Pass

•Additional Exercises – For Student's Practice

•*P(Pass | X) = P(X | Pass). P(Pass) / P(X)*

•*= P(X | Pass). P(Pass)*

•*= P(X | Pass). (0.44)*

•*= P(Male | Pass). P(3.5 | Pass).(0.44)*

•*= (2/4). 0.1056 . (0.44) = 0.0235*

•

$$p(X_i = x | Y = y_k) = \frac{1}{\sqrt{2\pi\sigma_{ik}^2}} \; e^{-\frac{1}{2}\left(\frac{x-\mu_{ik}}{\sigma_{ik}}\right)^2}$$

No of study hours	Gender	Final result
4.5	Male	Pass
7	Female	Pass
2	Male	Fail
4	Female	Fail
2.5	Male	Fail
3	Female	Fail
8.3	Male	Fail
8	Female	Pass
9	Male	Pass

•*P(Fail | X) = P(X | Fail). P(Fail) / P(X)*

•*= P(X | Fail). P(Fail)*

•*= P(X | Fail). (0.56)*

•*= P(Male | Fail). P(3.5 | Fail).(0.56)*

•*= (3/5). 0.1846 . (0.56) = 0.0615*

Text Classification using Naive Bayes Classifier

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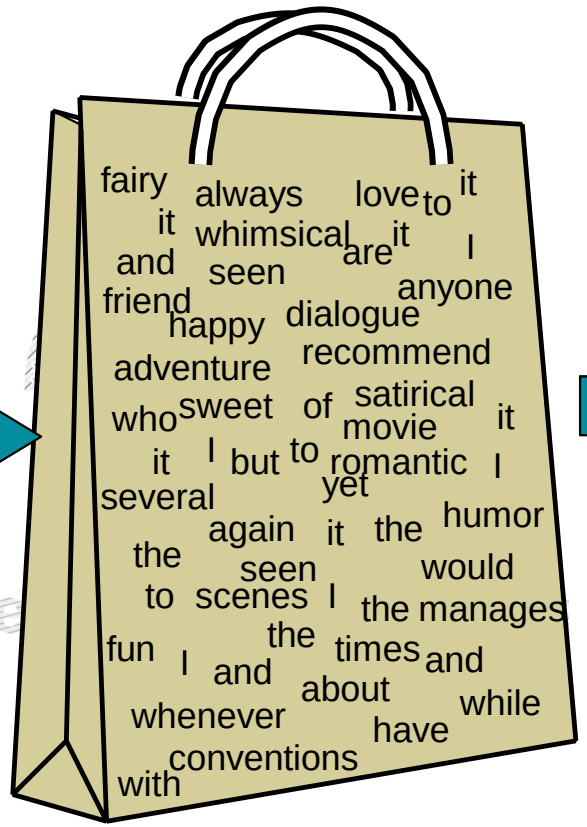
The Bag of Words Representation

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!



it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1

Example : Multinomial model :

Which Tag sentence " A very close game" belong to?

Text	Tag
"A great game"	Sports
"The election was over"	Not sports
"Very clean match"	Sports
"A clean but forgettable game"	Sports
"It was a close election"	Not sports

P(Sports=Yes) = 3/5
P(Sports=No) = 2/5
P(A | Sports=Yes) = 2/11
P(A | Sports=No) = 1/9

P(Sports=Yes | X) = P(X | Sports=Yes). P(Sports=Yes) / P(X)
= P(X | Sports=Yes). (3/5)
= P(A| Sports=Yes). P(Very | Sports=Yes). P(Close | Sports=Yes). P(Game | Sports=Yes). (3/5)
= (2/11) . (1/11) . (0/11) . (2/11) . (3/5)

A	Great	Game	The	Election	Was	Over	Very	Clean	Match	But	Forgettable	It	Close	*Sports or Not Sports*
1	1	1												1
			1	1	1	1								0
							1	1	1					1
1		1						1		1	1			1
1				1	1							1	1	0
1		1					1						1	????

Laplace Smoothing

- Laplace smoothing: we add 1 or in general constant k to every count so it's never zero.
- To balance this, we add the number of possible words to the divisor, so the division will never be greater than 1
- In our case, the possible words are ['a', 'great', 'very', 'over', 'it', 'but', 'game', 'election', 'clean', 'close', 'the', 'was', 'forgettable', 'match'].

- In our example
- we add 1 to every probability, therefore the probability, such as **P(close | sports)**, will never be 0.

Example : Multinomial model :

Which Tag sentence " A very close game" belong to?

Text	Tag
"A great game"	Sports
"The election was over"	Not sports
"Very clean match"	Sports
"A clean but forgettable game"	Sports
"It was a close election"	Not sports

P(Sports=Yes) = 3/5
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P(A | Sports=Yes) = 2/11
P(A | Sports=No) = 1/9

P(Sports=Yes | X) = P(X | Sports=Yes). P(Sports=Yes) / P(X)
= P(X | Sports=Yes). (3/5)
= P(A| Sports=Yes). P(Very | Sports=Yes). P(Close | Sports=Yes). P(Game | Sports=Yes). (3/5)
= (2/11) . (1/11) . (0/11) . (2/11) . (3/5)
= (2+1/11+14) . (1+1/11+14) . (0+1/11+14) . (2+1/11+14) . (3/5)
= 0.00002765

(3+14/5+28)
=0.00002396

A	Great	Game	The	Election	Was	Over	Very	Clean	Match	But	Forgettable	It	Close	*Sports or Not Sports*
1	1	1												1
			1	1	1	1								0
							1	1	1					1
1		1						1		1	1			1
1				1	1							1	1	0
1		1					1						1	????

Apply Laplace Smoothing

Word	P(word | Sports)	P(word | Not Sports)
a	2+1 / 11+14	1+1 / 9+14
very	1+1 / 11+14	0+1 / 9+14
close	0+1 / 11+14	1+1 / 9+14
game	2+1 / 11+14	0+1 / 9+14

$$P(a|Sports) \times P(very|Sports) \times P(close|Sports) \times P(game|Sports) \times P(Sports)$$
$$= 2.76 \times 10^{-5}$$
$$= 0.0000276$$

$$P(a|Not\,Sports) \times P(very|Not\,Sports) \times P(close|Not\,Sports) \times P(game|Not\,Sports) \times P(Not\,Sports)$$
$$= 0.572 \times 10^{-5}$$
$$= 0.00000572$$

Example 2: Multinomial model

	docID	words in document	in $c = China$?
training set	1	Chinese Beijing Chinese	yes
	2	Chinese Chinese Shanghai	yes
	3	Chinese Macao	yes
	4	Tokyo Japan Chinese	no
test set	5	Chinese Chinese Chinese Tokyo Japan	?

$$\hat{P}(w_t \mid C_k) = \frac{n_k(w_t)}{\sum_{s=1}^{|V|} n_k(w_s)},$$

N_{yes} (W=Chinese) = 5, N_{No} (W=Chinese) = 1,

|v| = 6 = {Chinese, Beijing, Shanghai, Macao, Tokyo Japan}
No of features (words) in Yes class = 8
No of features (words) in No class = 3

Source Credit : https://nlp.stanford.edu/IR-book

Example 2

	docID	words in document	in $c = China$?
training set	1	Chinese Beijing Chinese	yes
	2	Chinese Chinese Shanghai	yes
	3	Chinese Macao	yes
	4	Tokyo Japan Chinese	no
test set	5	Chinese Chinese Chinese Tokyo Japan	?

$$\hat{P}(w_t \mid C_k) = \frac{n_k(w_t)}{\sum_{s=1}^{|V|} n_k(w_s)},$$

Priors: $\hat{P}(c) = 3/4$ and $\hat{P}(\overline{c}) = 1/4$

$$\begin{aligned}
\hat{P}(\text{Chinese}|c) &= (5+1)/(8+6) = 6/14 = 3/7 \\
\hat{P}(\text{Tokyo}|c) = \hat{P}(\text{Japan}|c) &= (0+1)/(8+6) = 1/14 \\
\hat{P}(\text{Chinese}|\overline{c}) &= (1+1)/(3+6) = 2/9 \\
\hat{P}(\text{Tokyo}|\overline{c}) = \hat{P}(\text{Japan}|\overline{c}) &= (1+1)/(3+6) = 2/9
\end{aligned}$$

Source Credit : https://nlp.stanford.edu/IR-book

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Example 2

	docID	words in document	in $c = China$?
training set	1	Chinese Beijing Chinese	yes
	2	Chinese Chinese Shanghai	yes
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	4	Tokyo Japan Chinese	no
test set	5	Chinese Chinese Chinese Tokyo Japan	?

$$P(C_k|\mathcal{D}) \propto P(C_k) \prod_{j=1}^{\text{len}(\mathcal{D})} P(u_j|C_k)$$

u-each word in test document

$$\hat{P}(c|d_5) \propto 3/4 \cdot (3/7)^3 \cdot 1/14 \cdot 1/14 \approx 0.0003$$
$$\hat{P}(\overline{c}|d_5) \propto 1/4 \cdot (2/9)^3 \cdot 2/9 \cdot 2/9 \approx 0.0001$$

Source Credit : https://nlp.stanford.edu/IR-book

Naïve Bayes classifier: Summary Model

Model: joint probability distribution given by

- $P(X, Y) = P(Y)\ P(X|Y)$
- $P(X = X_1, \cdots, X_n, Y = y_k) = P(Y = y_k)\ P(X = X_1, \cdots, X_n | Y = y_k)$

Learning/Training:

For output variable Y

- $P(Y) \sim Ber(p)$

For each attribute X

- $P(X|Y) \sim \mathrm{Ber}(\pi)$ → Multivariate Bernoulli NB (X_i – binary)
- $P(X|Y) \sim \mathrm{Multinom}(\pi, n)$ → Multinomial NB (X_i – multinomial)
- $P(X|Y) \sim N(\mu, \sigma^2)$ → GaussianNB (X_i – real valued)

Source Credit : https://nlp.stanford.edu/IR-book

Logistic Regression vs Naïve Bayes

Idea:

- Naïve Bayes allows computing P(Y|X) by learning P(Y) and P(X|Y)

- Why not learn P(Y|X) directly?

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Logistic Regression and Gaussian Naïve Bayes Classifier

- Interestingly, the parametric form of P(Y|X) used by Logistic Regression is precisely the form implied by the assumptions of a Gaussian Naive Bayes classifier.

- Therefore, we can view Logistic Regression as a closely related alternative to GNB, though the two can produce different results in many cases

- Derivation given in CS-5 slides in self learning section and also attached PDF of new chapter from Tom Mitchell book

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Features of Bayesian learning

- Each observed training example can incrementally decrease or increase the estimated probability that a hypothesis is correct.

- Flexible approach to learning than algorithms that completely eliminate a hypothesis if it is found to be inconsistent with any single example.

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Practical Issues of Bayesian learning

- Require initial knowledge of many probabilities
  - Often estimated based on background knowledge, previously available data, and assumptions about the form of the underlying distributions.
- Significant computational cost required to determine the Bayes optimal hypothesis in the general case (linear in the number of candidate hypotheses)

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